Aadhaar Project: Interview Study Guide

Use this with the four notebooks open beside it.

How to use it: learn sections 1 to 6 first, then practise sections 7 to 9 out loud. Answer each question without reading, then check your answer.

1. Your pitches

30 seconds

> For the UIDAI Data Hackathon, we analysed 4.9 million Aadhaar enrolment and update records. When we audited our work, we found that most districts that appeared to be failing children had a record-keeping problem, not a service problem. New districts had their records split across old and new names. That changes the action from sending field teams to fixing the district master data.

2 minutes

> The problem was to find patterns and predictive indicators in Aadhaar enrolment and update data that could support decisions.
>
> Our first version had striking findings, such as a 5,000% surge and districts at zero compliance. When I rebuilt it from the raw data, several did not hold up. Row counts had jumped because the data changed from monthly snapshots to daily reporting. State rankings were wrong because district ratios had been added together. And missing data had been treated as zero.
>
> After fixing those, I looked at the districts that still seemed low. Using pincodes, I found that in 24 of 26 cases most child biometrics were filed under another district name, usually the older parent district. Their median share rose from 0.27 to 0.85.
>
> I then added anomaly detection using Isolation Forest and Local Outlier Factor, and an early-warning model. The anomaly methods agreed far above chance. The prediction models did not beat a simple rule, so I recommended the simple rule for now.
>
> My main recommendations are to fix the district master first, then focus on Manipur, West Bengal and Bihar for child updates, and Meghalaya and Nagaland for enrolment.

2. The project at a glance

Item	Detail
Problem	Find patterns, anomalies and predictive indicators to support decisions
Data	12 CSV files in three UIDAI datasets: enrolment, demographic updates, biometric updates
Extra data	UIDAI State/UT Aadhaar Saturation Report, 30 June 2026
Period	March to December 2025, with gaps
Tools	Python, pandas, scikit-learn, SciPy, matplotlib, Jupyter

Item	Detail
Notebooks	01 cleaning, 02 metrics, 03 anomalies, 04 early warning

Dataset columns

Dataset	Count columns
Enrolment	`age_0_5`, `age_5_17`, `age_18_greater`
Demographic	`demo_age_5_17`, `demo_age_17_`
Biometric	`bio_age_5_17`, `bio_age_17_`

All three also have `date`, `state`, `district` and `pincode`.

3. Suggested 15-minute presentation

Slides	Topic	Minutes	Main message
1	Problem and answer	1	Apparent failures were mostly a data problem
2	Data and scope	1	Three datasets, 4.9M rows
3	Data audit	2	Snapshots, then daily, and August is absent
4	Cleaning	1.5	4.9M to 4.3M rows, with repairs
5	Metric bugs	2	Why the old rankings were wrong
6	Pincode test	2	24 of 26 low districts explained
7	Anomaly detection	2	Methods agree, and flags were checked
8	Early warning	2	A simple rule matched ML
9	Recommendations	1	Fix data first, then target states
10	Limits and next steps	0.5	What the data cannot prove

4. Notebook walkthrough

Notebook 01: audit and cleaning

Purpose: create data that later steps can trust.

Step	Rows after	Removed
Raw data	4,938,837	
Valid dates and states	4,938,801	36
Valid pincodes	4,938,801	0
Valid districts	4,938,491	310
Remove duplicates	4,329,842	608,649

Final rows: biometric 1,761,911, demographic 1,591,929, enrolment 976,002.

The date audit is the key step. For each month, we counted rows, distinct dates and people.

Period	Reporting style
March to July	Mostly one date per month, with large counts per row
August	Absent from every file
September to December	Daily rows with small counts, and some gaps

Specific gaps:

- Biometric and demographic data stop on 20 September.
- 1 to 12 October is missing from all three datasets.
- December enrolment has only 11 dates.

Pincode repair: a pincode belongs to one state, so about 101,000 rows (2%) were moved to the most common state for their pincode. This fixed cases like Rupnagar appearing under Chandigarh.

Mapping bug fixed: the original mapping renamed every Aurangabad to Chhatrapati Sambhajinagar, and every Bijapur to Vijayapura. Bihar has its own Aurangabad, and Chhattisgarh has its own Bijapur, so these renames are now state-specific.

District names: state-district pairs fell from 1,077 to 907. This is still above India's official count, so district results need care.

Notebook 02: metrics and claim testing

Three calculation fixes

1. **Ratios:** add the counts first, then divide.

```
# Wrong: adding district ratios gives a score that grows with district count
state_df = district_df.groupby(['state', 'year_month']).sum(numeric_only=True)

# Right: sum the counts, then calculate the ratio
state_df = add_metrics(aggregate(biometric, demographic, enrolment, ['state']))
```

2. **Missing data:** the old code used `fillna(0)`, which turned "no data" into a real zero. Missing now stays missing.
3. **Time:** trend comparisons use only September to December, per reporting day.

Metric definitions

Metric	Formula	Meaning
Child biometric share	`bio_5_17 / (bio_5_17 + demo_5_17)`	Share of child updates that are biometric
Demographic share	demographic updates / all updates	Mix of update types
Adult enrolment share	`age_18_greater` / total enrolment	Share of new enrolments that are adults
Updates per 1,000	updates / population × 1,000	Workload per person

We renamed two metrics. "Compliance" became child biometric share, because the data has no count of children due. "Volatility" became demographic share, because the formula measures a mix.

Child share needs at least 30 child updates, since percentages on tiny counts are noise.

The pincode test

1. Take each district below 50%.
2. Find its pincodes in the demographic file.
3. See which district name the biometric file gives for those same pincodes.
4. Recalculate the share across both name versions.

Example: Didwana-Kuchaman had 1 child biometric under its own name and 26,290 under Nagaur.

Result	Value
Districts tested	26
Most biometrics under another name	24
Median share before	0.27
Median share by pincode	0.85
Districts rising to 70% or above	20

Notebook 03: anomaly detection

- **Data:** 842 districts with enough activity.
- **Features:** child biometric share, demographic share, adult enrolment share, update intensity, peak-day share and weekend share.
- **Scaling:** RobustScaler, because outliers would distort mean-based scaling.
- **Size:** it is not a feature, so districts are not flagged just for being big.

Check	Result
Flagged by Isolation Forest and LOF	43 each
Flagged by both	25, against about 2 by chance
Score rank correlation	0.76
Flags surviving 20 random seeds	95% on average
Correlation of score with size	-0.41
Flagged by at least two methods	43
Explained by naming issue	13
Largest state cluster	Meghalaya, 11 districts

Date check: each date is compared with the median for that weekday, then scored with a robust z-score. Of 82 dates, 10 were flagged, including 20 to 23 October, around Diwali. Some flags sit beside data gaps, so they need checking.

Notebook 04: early-warning model

Question: will this district be in the bottom quarter for child biometric share next month?

Features: current child share, state child share, demographic share, adult enrolment share, and logged child updates, total updates and enrolment.

Split	Data
Train	September to October, and October to November
Test	November to December

Approach	ROC AUC	Without naming districts	Precision in top 10%	Recall in top 25%
Persistence	0.901	0.892	0.94	0.735
Random Forest	0.899	0.889	0.94	0.735
Logistic Regression	0.893	0.883	0.94	0.755

- Base rate of high risk: about 24.5%.
- Logistic minus persistence, 95% interval: -0.021 to 0.004. This includes zero, so there is no reliable difference.
- Strongest predictor: current child share (coefficient -1.48).
- Random Forest: 400 trees, minimum 5 samples per leaf.

5. Concepts in plain English

Concept	Simple explanation	Why it was used
Z-score	How many standard deviations a value is from the mean	Simple baseline
Robust z-score	Uses median and spread around it, so outliers matter less	Daily data has extreme days
Isolation Forest	Random splits isolate unusual points quickly	Finds unusual combinations
LOF	Compares a point's local density with its neighbours	Finds points odd for their peer group
Contamination	The share of points the model is told to flag	Set to 5% for review capacity
RobustScaler	Scales using median and interquartile range	Resistant to outliers
Persistence	Assume this month's pattern continues	Baseline a model must beat
Logistic Regression	Linear model that gives a probability	Easy to interpret
Random Forest	Averages many decision trees	Captures non-linear patterns
ROC AUC	Chance a random positive ranks above a random negative	Measures ranking quality
PR AUC	Precision and recall across thresholds	Focuses on the positive class
Precision in top 10%	Of the flagged list, how many are correct	Teams review limited lists
Recall in top 25%	How many true cases the list catches	Measures coverage

Concept	Simple explanation	Why it was used
Bootstrap	Resample the test set many times	Shows how much a score may vary by chance
Spearman correlation	Correlation between ranks	Handles skew and non-linearity
p-value	How surprising the result would be if there were no real link	Here 0.20, so not significant
Data leakage	Using information unavailable at prediction time	Prevented with a time-based split

6. Findings to remember

#	Finding	Evidence	Action
1	Low districts are mostly a naming issue	24 of 26, median 0.27 to 0.85	Fix the district master
2	The data format changed	Rows about 50×, August absent	Publish a data dictionary
3	Manipur, West Bengal and Bihar have the lowest child share	About 74 to 75%, against UP 84%	Review child updates there
4	Updates dominate	62% biometric, 33% demographic, 4.8% enrolment	Plan capacity around updates
5	Meghalaya and Nagaland stand out	Meghalaya 37% adult, 84% saturation. Nagaland 62% saturation	Targeted enrolment
6	Workload per person varies	Manipur 102 per 1,000, median 31	Use volume and per-person load
7	Maharashtra's November rise was real	Thane and Pune about doubled	Plan for demand spikes
8	Low-share districts stay low	Persistence 0.901	Use a simple watchlist for now

What changed from the first report

First report	Rebuilt result
UP lowest compliance	16th lowest of 36
60+ districts at 0%, 7.5M children	No district with enough data at 0%
5,000% September surge	Compares snapshots with daily data
Thane and Pune 7×	About 2×
Major states peaked in November	November busiest in 4 states, September in 22
Manipur 45× median, 15% biometric	3.3× median, 38% biometric
Top 5 states nearly 40%	51.6% of updates
ML models deployed	No such code existed, now built and tested
390,612 duplicates	608,649 duplicates

7. Expected questions and answers

Project and problem

Why did you choose this project? It is a real public dataset with messy data and a clear decision at the end. It also let me show how checking the work changed the conclusion.

What was the business problem? UIDAI needs to know where to direct operational effort. The risk is acting on misleading signals and sending resources to the wrong place.

What are you most proud of? The pincode test. It was simple, and it changed the main recommendation.

What was the hardest part? Accepting that some exciting findings were wrong, then working out exactly why.

What would you do differently? Audit the dates and check ratios before building any insight.

Data and cleaning

How did you start? With a data audit: counts by month, distinct dates, duplicates, and state and district validity.

How did you discover the format change? Rows jumped far more than people counted, so I checked distinct dates per month. March to July mostly had a single date.

Why remove duplicates last? Different spellings can hide duplicates until names are standardised.

Can duplicates be real records? Exact matches on date, location and every count are very likely repeats, but with aggregated data it isn't certain. That is an assumption I would confirm with the data owner.

How did you handle missing data? I kept it missing. Zero means nothing happened, while missing means we don't know.

Why is 907 still more than India's official districts? Some name variants and split districts remain. I'd fix this with an official district master such as the Local Government Directory.

Why use a minimum of 30 child updates? Two updates out of two gives 100%, which means little. The threshold is a judgement call to reduce noise.

Metrics and analysis

Explain the ratio bug. State scores were built by adding district ratios, so states with more districts scored higher. The old score correlated 0.98 with district count. The fix is to sum counts first, then divide.

Why is it not called compliance? Compliance needs the number of children due for an update, which the data lacks. Our metric only shows the mix of child updates.

How would you measure true compliance? Divide completed mandatory updates by children reaching the required ages, using cohort data from UIDAI.

Why report per reporting day? Months have different numbers of dates, so raw monthly totals aren't comparable.

Why September to December only? It is the only period with daily data in the same format.

Does high adult enrolment mean low saturation? Not across all states. The correlation was -0.22 with $p = 0.20$. Meghalaya is a clear individual case, but that isn't a national rule.

Why does saturation exceed 100% in some states? Population is a projection, and the count depends on current address, including people who have moved. The report notes that holders may not be domiciled in

that state.

Pincode test

Is this proof? No, it's strong evidence. Pincodes can cross district borders, and 2 of the 26 weren't explained. I would confirm with UIDAI's district master.

Why does this happen? My likely explanation is that one dataset moved to new district names sooner than the other. The test shows the pattern but not the cause.

How does that change the action? Those districts need a data fix rather than outreach, which saves field resources.

Could the corrected 0.85 be overstated? It covers whole pincodes, which may include neighbouring areas, so treat it as an estimate. My first version overstated it at 0.97 because the two sides weren't like for like, and I corrected that.

Anomaly detection

Why unsupervised? There are no labels telling us which districts are truly anomalous.

How did you validate it? Agreement between methods, stability across seeds, a size check, and manual review of flags.

Isolation Forest or LOF: what's the difference? Isolation Forest finds points that are globally easy to separate. LOF finds points that are unusual compared with nearby points.

Why flag 5%? It reflects review capacity. It's a business choice and doesn't mean 5% of districts are truly anomalous.

Why are flags biased towards small districts? Small districts have noisier percentages. That's why a flag is a lead to review, not a verdict.

Why was size excluded? Otherwise the models would mostly flag large districts for being large.

Why not use the z-score alone? It checks one metric at a time and misses unusual combinations.

Prediction

Why a time-based split? In real use, you predict the future from the past. A random split could leak future patterns.

Why not cross-validation? Standard k-fold mixes time periods. With so few months, one forward test was the cleanest option.

Why use persistence as a baseline? If a simple rule works, a complex model must justify itself.

Why didn't ML win? The current share already contains most of the signal, and there are only three month pairs of history.

So was ML useless? No. It tested whether extra complexity adds value and showed it doesn't yet. That saves building and maintaining something unnecessary.

Does the naming issue inflate performance? I removed those 18 test districts and AUC stayed around 0.89.

Are the risk scores probabilities? Treat them as rankings only. With so little data they aren't calibrated.

How would you improve the model? Add more months, a proper eligible-children count, centre capacity, and location data.

Why not XGBoost? It might match Random Forest, but with this little history I'd expect no real gain.

Why were LSTM and Prophet dropped? About 90 days with gaps can't reliably train them. A weekday baseline answers the same question more simply.

Business and impact

What would you tell UIDAI's leadership? Before sending field teams, fix the district records. Then focus child outreach on Manipur, West Bengal and Bihar, and enrolment on Meghalaya and Nagaland.

How would you measure success? Fewer unexplained low-share districts, a higher true update rate among children due, and shorter update backlogs.

How would you put this into production? A scheduled pipeline loads new data, runs quality checks, updates the metrics and publishes a watchlist to a dashboard, with data-quality alerts.

How does this connect to ZS? Pharma analytics has the same issues: messy geography, changing territories, and acting on misleading signals. A territory realignment could create the same split between old and new names. Validating data before a decision applies directly.

Your role and integrity

What was your contribution? *Prepare your honest answer. Be specific about what you did in the team, and what you did in the rebuild.*

The original report had errors. Why? We moved fast and trusted the outputs before auditing them. I went back to the raw data, found the causes and corrected them.

Did the first report really use those ML models? *Check with your team first.* If there was no such code, say: "I couldn't find that code in our notebooks, so I rebuilt the anomaly work properly and dropped the rest."

Did you use AI? Yes, and say so honestly. For example: "I used AI assistance for the rebuild and code, then reviewed the logic and results myself." Only say this if it's true, so study the notebooks until it is.

Why should we trust your numbers? They reconcile back to the raw counts, the notebooks run from start to finish, and the main claims were tested against the data.

8. Likely code questions

Be ready to explain these lines.

```
# Outer merge keeps districts missing from one dataset
enrol_dist.merge(bio_dist, on=keys, how='outer')
```

```
# Keeps the share only where there is enough data
(bio / child_updates).where(child_updates >= MIN_CHILD_UPDATES)
```

```
# Isolation Forest: -1 means anomaly
isolation = IsolationForest(n_estimators=300, contamination=0.05, random_state=7).fit(X)
usable['flag_iforest'] = isolation.predict(X) == -1
```

```
# Local Outlier Factor
lof = LocalOutlierFactor(n_neighbors=20, contamination=0.05)
usable['flag_lof'] = lof.fit_predict(X) == -1
```

```
# Time-based split
train = pd.concat([pairs[(months[0], months[1])], pairs[(months[1], months[2])]])
test = pairs[(months[2], months[3])]
```

```
# Persistence: lower current share means higher risk
scores = {'Persistence': -X_test['child_biometric_share'].to_numpy()}
```

Practise these changes: the 30-update threshold, 5% flag rate, or the bottom-quarter target. Explain what would happen to the results.

9. Other parts of the round

Online test questions

This project cannot help here. Rework every question you remember. Common areas to revise are:

- **SQL:** joins, `GROUP BY`, window functions, ranking.
- **Python:** pandas, loops, functions, debugging.
- **Statistics:** mean and median, probability, hypothesis tests, p-values.
- **ML:** overfitting, regularisation, bias and variance, evaluation metrics.

Business case framework

Use this order every time:

1. **Objective:** what decision is being made?
2. **Metric:** how will success be measured?
3. **Data:** what's needed, and what could go wrong?
4. **Baseline:** what's the simple answer?
5. **Approach:** which method fits, and why?
6. **Validation:** how will you test it?
7. **Deployment:** who uses it, and how?
8. **Risks:** data quality, bias, change management.

Practice case: which doctors should sales reps visit?

- **Objective:** improve prescribing outcomes with limited rep time.
- **Metric:** incremental prescriptions, not total prescriptions.
- **Data:** prescriptions, call history, specialty, geography, payer access.
- **Baseline:** rank by current prescription volume.
- **Approach:** segment doctors, then predict response to calls.
- **Validation:** time-based holdout, ideally a controlled pilot.

- **Deployment:** a call plan in the sales system, refreshed regularly.
- **Risk:** high prescribers may not need visits, so measure response, not volume.

Practice case: reduce Aadhaar centre wait times

- **Objective:** lower waits within the current budget.
- **Metric:** average and peak wait, completed updates per day.
- **Data:** daily volumes, staff, kits, hours, holidays, deadlines.
- **Baseline:** staffing based on the same weekday's average.
- **Approach:** forecast demand, then allocate staff and kits with optimisation.
- **Validation:** backtest, then pilot in a few districts.
- **Deployment:** a weekly staffing plan.
- **Risk:** deadlines and campaigns create sudden spikes.

Likely technology answer

For a real system: **SQL or a cloud warehouse** for storage, **Python** for analysis, **Spark** if data is very large, **Airflow** for scheduling, **MLflow** for tracking models, and **Power BI or Tableau** for reporting. Choose only what the problem needs.

10. Claims to handle carefully

Say this	Avoid this
Child biometric share	Compliance rate
Most likely a naming issue	Proven naming error
Pattern is consistent with	This caused
Scores rank risk	99% chance of failure
The data cannot test the fee effect	The fee hike caused the cliff
About 2×	The old 7× or 5,000% figures
Probable reporting change	Definite system migration

Things you should not claim:

- That districts truly have zero child biometrics.
- That 7.5 million children are affected.
- That fraud or bots were detected.
- That the data proves why any spike happened.
- That the project uses LSTM, Prophet or graph models.

Limits to state up front:

- The metric isn't true compliance.
- Trend work covers only September to December, with gaps.
- The pincode test is strong evidence, not proof.
- There is only one test month.

- Population figures are 2026 projections.

11. Numbers to memorise

Topic	Number
Raw rows	4,938,837
Clean rows	4,329,842
Duplicates removed	608,649
State-district pairs	1,077 to 907
Rows jump in September	About 50×
Old score vs district count	0.98 correlation
UP's real rank	16th lowest of 36
Low districts explained	24 of 26
Median share	0.27 to 0.85
Update mix	62% / 33% / 4.8%
Top five states	51.6% of updates
Manipur	102 per 1,000, 3.3× median
Meghalaya	37% adult, 84% saturation
Nagaland saturation	62%
Anomaly overlap	25, against about 2 by chance
Seed stability	95%
Persistence / RF / LR	0.901 / 0.899 / 0.893

12. Final checklist

Before the interview

- ☐ Say the 30-second and 2-minute pitches without notes.
- ☐ Run all four notebooks yourself.
- ☐ Explain every code snippet in section 8.
- ☐ Explain every concept in section 5.
- ☐ Prepare your honest answer on your role and team credit.
- ☐ Confirm with your team whether any ML code exists elsewhere.
- ☐ Build the slides and stop using the old report.
- ☐ Rework your online test questions.
- ☐ Practise two business cases out loud.

On the day

- ☐ Keep notebooks, charts and slides available offline.

- ☐ Have the README open as a backup.
- ☐ Start with the answer, then explain how you reached it.